Baderia Global Institute of Engineering & Management
Global Square, Patan Bypass, Raigwan, Jabalpur (M.P.)
Approved by AICTE, New Delhi | Affiliated to RGPV, Bhopal

Date: _____

To,

The Principal,

_____

_____

Subject: Proposal for "Code the Voice", a three-day AI Voice Agent Bootcamp for students of Classes _____ to _____

Respected Sir/Madam,

Baderia Global Institute of Engineering & Management (BGIEM), Jabalpur, is pleased to propose **Code the Voice**, a three-day hands-on bootcamp in which students learn to build AI voice agents.

Voice AI agents now answer calls for banks, hospitals and customer service teams. In this bootcamp, students learn how these systems work and build them with the tools the industry uses: ElevenLabs for no-code agents, and Pipecat, an open-source Python framework, for code-based agents. Each team builds an agent that holds a natural conversation, answers from a knowledge base, books appointments and follows safety guardrails, and presents it live on the final day.

The bootcamp runs for three hours a day over three days in the school's computer lab. It is conducted by our trainers free of cost, as part of the Institute's school outreach programme.

The detailed proposal is enclosed for your kind consideration. We would be glad to discuss it with you and to fix dates that suit the school.

Thanking you,

Yours sincerely,

Dr. Niharika Yadav
Deputy Director
Baderia Global Institute of Engineering & Management, Jabalpur

Enclosure: Detailed proposal for the Code the Voice bootcamp

WORKSHOP PROPOSAL

CODE THE VOICE

The 3-Day AI Voice Agent Bootcamp

Build an AI that listens, thinks and talks back.

Siri, Alexa and Google Assistant were built by large teams of engineers. In three days, your students will build their own voice agent.

Every team builds it twice: first on ElevenLabs, the no-code platform that businesses use to launch voice agents, and then in Python with Pipecat, the open-source framework engineers use in production. The finished agent holds a natural conversation, answers from its own knowledge base and books appointments on a calendar. On the last day, it takes a live call in front of the school's teachers.

3	540	4	1
days	minutes of learning	modules	live voice agent per team

1. THE BOOTCAMP AT A GLANCE

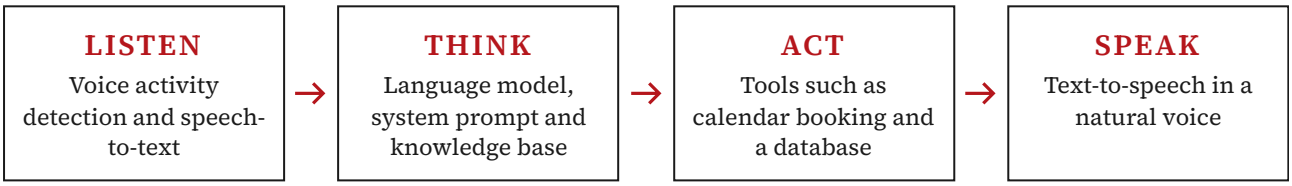
Programme	Code the Voice: The 3-Day AI Voice Agent Bootcamp
Proposed for	_____
Participants	Students of Classes _____ to _____; up to 40 students per batch, working in teams of 2 or 3
Duration	3 days, 3 hours a day (540 minutes in all). The days can be consecutive or spread over one or two weeks.
Venue	School computer lab
Format	Concepts, live demonstrations and guided building, with most of the time spent on hands-on work
Tools	ElevenLabs Agents platform (no-code) and Pipecat, the open-source Python framework (code)
Prerequisite	Basic programming, preferably in Python. No prior knowledge of AI is needed.
Setup	BGIEM prepares the accounts and software with the school's IT team before Day 1.
Lead trainer	Mr. Deepak Vishwakarma, Lead Corporate Trainer, BGIEM
Cost to the school	Nil

2. WHY THIS BOOTCAMP

- **Voice AI is where the industry is heading.** Banks, hospitals, recruiters and logistics companies now use AI agents to answer and make calls. Students learn the technology behind them.
- **Students use real industry tools.** ElevenLabs and Pipecat are used to build voice agents for actual businesses, not just for classrooms.
- **It covers both paths.** Students first build without code to understand design, then build in code to understand what happens underneath.
- **It stretches students who can already code.** The finished agent is a strong project for science exhibitions, hackathons and college applications.

3. WHAT STUDENTS WILL BUILD

Every team builds a voice agent on the cascade architecture, the design used in most commercial voice agents:



Students build this pipeline twice: once on ElevenLabs without code, and once in Python with Pipecat.

Each team chooses a real use for its agent, for example a **school helpdesk**, a **clinic appointment desk**, a **college admission enquiry line** or an **exam revision tutor**.

4. COURSE STRUCTURE

Module	Title and topics	Day	Minutes
1	**AI Voice Agents Landscape** Bootcamp roadmap; voice AI use cases and return on investment; why voice AI skills matter	Day 1	45
2	**Foundations of AI Voice Agents and Their Architectures** Inbound and outbound agents; cascade architecture; speech-to-speech architecture; build paths, challenges and what's next	Day 1	115
3	**Building Voice Agents on a No-Code Platform (ElevenLabs)** Getting started; prompt, voice and LLM setup; workflows and sub-agents; A/B testing and knowledge base; calendar booking and system tools; guardrails; advanced settings; live testing	Day 2	160
4	**Building Code-Based Voice Agents with Pipecat** Why code-based agents; live demo of a booking agent; building the Pipecat pipeline; pipeline recap	Day 3	100
–	**Capstone, Demo Day and Feedback** Final polish of each team's agent, live demonstration and feedback	Day 3	70

Module	Title and topics	Day	Minutes
–	Breaks, recaps and daily wrap-ups	All	50
	Total		**540**

5. DAY-WISE TIMING BREAKDOWN

DAY 1

Understand the Voice AI World

Module 1 and Module 2

Time	Min.	Session	What students do
0:00 – 0:15	15	**Kick-off and Roadmap**	A finished voice agent takes a live call in front of the class. Overview of the three days and of what each team will build.
0:15 – 0:35	20	**Use Cases and ROI**	How banks, hospitals, recruiters and logistics companies use voice agents, and why: service round the clock, scale and lower cost per call.
0:35 – 0:45	10	**Why Voice AI Skills Matter**	Jobs and skills in voice AI, and where the field is heading.
0:45 – 1:00	15	**Inbound and Outbound Agents**	Agents that answer calls and agents that make them, with examples of each. Teams decide which kind they will build.
1:00 – 1:10	*10*	*Break*	
1:10 – 2:10	60	**Cascade Architecture**	Telephony, voice activity detection and turn-taking, speech-to-text, the language model, system prompt, knowledge base, tools, memory, streaming, interruptions and transfer to a human. Mini-lab: each team builds a simple listen–think–speak loop in the browser.
2:10 – 2:30	20	**Speech-to-Speech Architecture**	How newer models listen and speak directly. Comparison with the cascade on delay, emotion, reliability of tools and monitoring, with a live demonstration.
2:30 – 2:50	20	**Build Paths and Challenges**	No-code platforms and code frameworks compared. Real problems such as delay, background noise, turn-taking and safety, and what comes next.
2:50 – 3:00	10	**Day Wrap-up**	Quick quiz on the day's concepts. Teams choose the use case for their agent.
Day 1 total	**180**	**3 hours**	

Day 1 output: students understand how voice agents work, have built a basic voice loop and have chosen the use case for their agent.

DAY 2

Build It Without Code

Module 3: ElevenLabs

Time	Min.	Session	What students do
0:00 – 0:10	10	**Recap**	Quick recap of Day 1. Teams confirm their use case.
0:10 – 0:30	20	**Getting Started with ElevenLabs**	A tour of the agents platform. Each team creates its first agent and makes a test call.
0:30 – 1:15	45	**Prompt, Voice and LLM Setup**	Writing a structured system prompt covering the agent's role, goal, tone, steps and rules. Choosing a voice and a language model, writing the first message, and setting up English and Hindi.
1:15 – 1:25	*10*	*Break*	
1:25 – 1:50	25	**Workflows and Sub-Agents**	Designing the conversation as a flow, and handing over between specialist sub-agents.
1:50 – 2:05	15	**A/B Testing and Knowledge Base**	Adding a knowledge base document so that the agent answers from facts, and comparing two versions of the agent to see which works better.
2:05 – 2:25	20	**Calendar Booking and System Tools**	Connecting the agent to a demo Google Calendar so that it can book appointments. System tools such as ending a call and transferring to a human.
2:25 – 2:35	10	**Guardrails**	Keeping the agent on its topic and protecting it from off-topic and manipulative conversations.
2:35 – 2:45	10	**Advanced Settings**	Turn-taking, interruptions, response speed and voice settings.
2:45 – 3:00	15	**Live Testing**	Teams call each other's agents, try to break them and note what to fix.
Day 2 total	**180**	**3 hours**	

Day 2 output: a working no-code voice agent with a knowledge base, calendar booking and guardrails.

DAY 3 **Build It in Code**

Module 4: Pipecat, Demo Day and Feedback

Time	Min.	Session	What students do
0:00 – 0:15	15	**Why Code-Based Voice Agents**	When code is the better choice: full control, custom logic, lower running cost and open-source tools.
0:15 – 0:30	15	**Live Demo: Booking a Session**	A code-based agent built with Pipecat books a session on a call, from start to finish.
0:30 – 1:30	60	**Building the Pipecat Pipeline**	Guided coding in Python: frames and pipelines, WebRTC transport, Silero voice activity detection, speech-to-text, the language model with conversation context, text-to-speech, interruptions, and function calling with a booking database.
1:30 – 1:40	*10*	*Break*	
1:40 – 1:50	10	**Pipeline Recap**	The full pipeline reviewed step by step, and the no-code and code approaches compared side by side.
1:50 – 2:10	20	**Capstone Polish**	Teams finalise their agent's prompt, knowledge base, tools and guardrails for Demo Day.
2:10 – 2:45	35	**Demo Day**	Each team's agent takes a live call in front of a panel of the school's teachers.
2:45 – 3:00	15	**Feedback and Closing**	Feedback form, certificates and awards.
Day 3 total	**180**	**3 hours**	

Day 3 output: a finished voice agent presented live, and a code-based pipeline that students understand line by line.

Total time	Minutes	Hours
Day 1: Modules 1 and 2	180	3
Day 2: Module 3	180	3
Day 3: Module 4, Demo Day and Feedback	180	3
Total	**540**	**9**

6. LEARNING OUTCOMES

By the end of the bootcamp, students will be able to:

1. explain where voice AI agents are used and why businesses are adopting them;
2. describe the cascade and speech-to-speech architectures and choose between them;
3. build a no-code voice agent with a structured prompt, knowledge base, workflows, tools and guardrails;
4. build a code-based voice pipeline in Python with Pipecat, including function calling;

5. deal with real-world problems such as delay, turn-taking, interruptions and background noise;
6. test an agent, improve it and present it as a product.

7. WHAT STUDENTS TAKE HOME

- The design of their no-code agent: its system prompt, workflow and knowledge base, so that they can build it again later.
- The complete Pipecat project code, with comments.
- A bootcamp handbook that explains every module step by step.
- A certificate of completion from BGIEM. The top three teams also receive awards.
- Four weeks of online doubt support through a Google Classroom managed by the school.

8. STUDENT SAFETY AND DATA PRIVACY

- No student creates an account on any AI service. Teams work in workspaces managed by BGIEM.
- The keys used in the Pipecat module are temporary and limited, and are switched off after the bootcamp.
- Every agent has guardrails, so that it stays on its topic and refuses unsafe requests.
- Test calls use sample details only. Calendar booking uses a BGIEM demo calendar and a sample database, never real personal data.
- Photographs are taken only as per the school's policy.

9. COST TO THE SCHOOL

The bootcamp is offered free of cost. BGIEM bears all expenses, including the trainers, platform accounts, materials, certificates and awards. The school only provides its computer lab for the three days.

10. ABOUT THE INSTITUTE

Baderia Global Institute of Engineering & Management, Jabalpur, was established in 2009. It is approved by AICTE and affiliated to Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV), Bhopal, and offers programmes in engineering and management, including B.Tech in Computer Science and Engineering with specialisation in Artificial Intelligence and Machine Learning. Through its school outreach programme, the Institute brings practical technology workshops to school students.

Mr. Deepak Vishwakarma
Lead Corporate Trainer, BGIEM (Prepared by)

Dr. Niharika Yadav
Deputy Director, BGIEM (Approved by)